

Convergence Tests

ANSWER KEY



The ratio and comparison tests

- 1 Use the ratio test to determine whether $\sum_{n=1}^{\infty} \frac{n}{2^n}$ converges.
 $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \frac{n+1}{2n} = \frac{1}{2} < 1$: converges.
- 2 Use the ratio test on $\sum_{n=1}^{\infty} \frac{n!}{n^n}$.
 $\frac{a_{n+1}}{a_n} = \frac{n^n}{(n+1)^n} = \left(\frac{n}{n+1}\right)^n \rightarrow \frac{1}{e} < 1$: converges.
- 3 Use the comparison test to determine whether $\sum_{n=1}^{\infty} \frac{1}{n^2+3}$ converges, by comparing to $\sum_{n=1}^{\infty} \frac{1}{n^2}$.
 $0 < \frac{1}{n^2+3} < \frac{1}{n^2}$ for all $n \geq 1$, and $\sum_{n=1}^{\infty} \frac{1}{n^2}$ is a convergent p -series ($p = 2 > 1$), so by comparison the given series converges.

Alternating series and the p-series test

- 4 Determine whether each p -series converges or diverges:
a $\sum_{n=1}^{\infty} \frac{1}{n^3}$ Converges ($p = 3 > 1$). **b** $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$ Diverges ($p = \frac{1}{2} \leq 1$).
- 5 Determine whether $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$ converges, and classify it as absolutely convergent, conditionally convergent, or divergent.
The terms $\frac{1}{n}$ decrease to 0, so by the alternating series test the series converges. But $\sum_{n=1}^{\infty} \frac{1}{n}$ (absolute values) diverges — so the series is conditionally convergent.